

JUPITER'S MOONS

Jupiter's moons fall into three categories: the four inner moons; the four large Galilean moons; and the rest, the small outer moons.

The inner and Galilean moons orbit in the usual direction—that is, the same direction as Jupiter's spin (counterclockwise viewed from

above the north pole). Most of the outer moons travel in the opposite direction, suggesting that they originated from an asteroid that fragmented after it was captured by Jupiter's gravitational field.

SO NEAR AND YET SO FAR

Io, one of the largest of Jupiter's 64 moons, appears close to its planet, but the two are almost three times the diameter of Jupiter apart.

INNER MOON

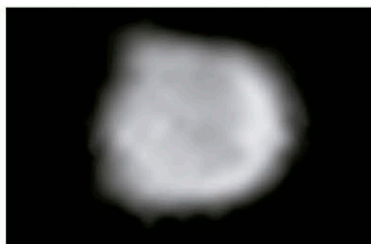
Metis

DISTANCE FROM JUPITER 79,460 miles (127,960 km)

ORBITAL PERIOD 6 hours 58 minutes

DIAMETER 25 miles (40 km)

Metis, the closest moon to Jupiter, is irregular in shape and lies within the planet's main ring. It was discovered on March 4, 1979, by the Voyager 1 probe. Metis is named after the first wife of Zeus, who was swallowed by him when she became pregnant.



JUPITER'S CLOSEST MOON

INNER MOON

Adrastea

DISTANCE FROM JUPITER 80,100 miles (128,980 km)

ORBITAL PERIOD 7 hours 9 minutes

LENGTH 16 miles (26 km)

The small, irregularly shaped Adrastea is the second moon out from Jupiter and lies within its main ring. For each orbit of Jupiter, Adrastea spins once on its axis, so the same side of the moon always faces the planet. This synchronous rotation is also exhibited by Adrastea's three closest neighbors, Metis, Amalthea, and Thebe. Adrastea was discovered by Voyager 2 in July 1979, and is named after a nymph of Crete into whose care, according to Greek mythology, the infant Zeus was entrusted.



ADRASTEA

INNER MOON

Thebe

DISTANCE FROM JUPITER 137,800 miles (221,900 km)

ORBITAL PERIOD 16 hours 5 minutes

LENGTH 68 miles (110 km)

The most distant of the inner moons, Thebe is named after an Egyptian king's daughter who was a granddaughter of Io. The moon, which was discovered on March 5, 1979, by Voyager 1, lies within the outer part of the Gossamer Ring (see p.181).



THEBE SHOWING IMPACT CRATER

INNER MOON

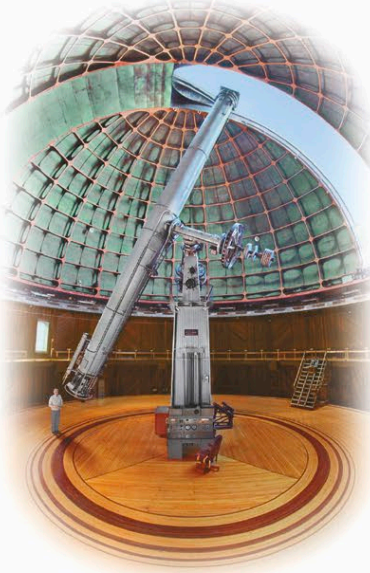
Amalthea

DISTANCE FROM JUPITER 112,590 miles (181,300 km)

ORBITAL PERIOD 11 hours 46 minutes

LENGTH 163 miles (262 km)

The largest of Jupiter's inner moons and the third from the planet, Amalthea is named after the nurse of newborn Zeus. The irregularly shaped moon lies within the Gossamer Ring and is believed to be a source of ring material. Meteoroids from outside the Jovian system collide with Amalthea and the other inner moons, chipping off flecks of dust, which then become part of the ring system. Amalthea's unexpected discovery on September 9, 1892, over 280 years after the four much larger Galilean moons had been discovered was headline news.

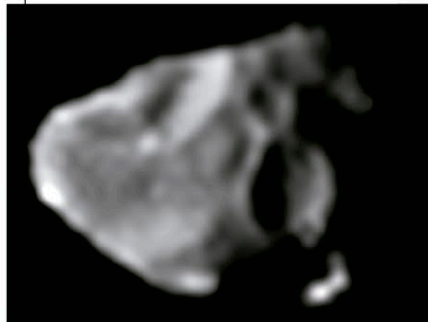


BARNARD'S TELESCOPE

Amalthea was the last of Jupiter's moons to be discovered by direct visual observation (as opposed to photography). Its discoverer, American Edward Barnard, used a 36-in (91-cm) refractor telescope, which is now preserved at the Lick Observatory in California.

BATTERED SURFACE

The circular feature in this image is Pan, which, with a diameter of about 56 miles (90 km), is the largest impact crater on Amalthea. The bright spot below Pan is associated with another, smaller crater, Gaea (bottom).



GALILEAN MOON

Europa

DISTANCE FROM JUPITER 416,630 miles (670,900 km)

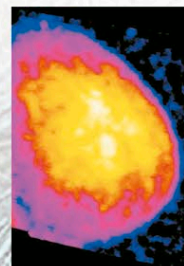
ORBITAL PERIOD 3.55 Earth days

DIAMETER 1,939 miles (3,122 km)

Europa is an ice-covered ball of rock, which has been studied for about 400 years but whose intriguing nature was only fully revealed once the Galileo space probe started its study in 1996. The probe was named after the Italian scientist Galileo Galilei, who observed Europa, along with the three other moons that collectively bear his name, in January 1610, from Padua, Italy. The German astronomer Simon Marius (1573–1624) is believed to

DAYTIME TEMPERATURE

Infrared observations reveal heat radiation from Europa's surface at midday. Temperatures at the equator (shown here as yellow) reach about -225°F (-140°C). Farther away from the equator, the surface temperatures are even lower.



OUTER MOON

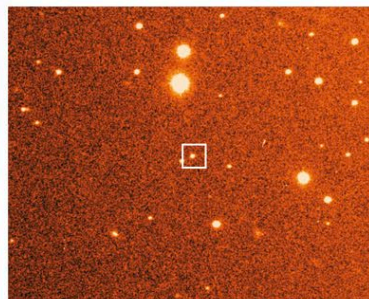
Themisto

DISTANCE FROM JUPITER 4.66 million miles (7.5 million km)

ORBITAL PERIOD 130 Earth days

DIAMETER 5 miles (8 km)

In November 2000, astronomers at the Mauna Kea Observatory, Hawaii, carried out a systematic search for new moons and identified 11 small moons. Observations recorded on subsequent nights revealed that one of the 11, since named Themisto, was a moon that had been discovered by American astronomer Charles Kowal on September 30, 1975, but then lost.



THEMISTO REDISCOVERED

This digital image is one of a series that shows Themisto (highlighted) and its changing position against the background stars, which led to its rediscovery in November 2000.